

Exact Half-Chromatic Numbers of Continuous Complete Graphs

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Abstract

Grigoriev and Faulkner introduced colouring problems on continuous graphs and conjectured that the radius- $\frac{1}{2}$ chromatic number of the continuous complete graph satisfies

$$\chi_{\frac{1}{2}}(\Gamma(K_n)) = \left\lceil \frac{n}{2} \right\rceil.$$

We determine the exact value for every $n \geq 5$ and thereby refute the conjecture. Define

$$\lambda(n) = \min \left\{ c \in \mathbb{Z} : 3 \leq c \leq n, \binom{n-c}{2} \leq c(c-2) \right\}.$$

Then

$$\chi_{\frac{1}{2}}(\Gamma(K_n)) = \lambda(n) = \left\lceil \frac{\sqrt{8n^2 - 24n + 25} - (2n - 5)}{2} \right\rceil \quad (n \geq 5).$$

The lower bound is obtained by orienting the support edge of every interior centre toward a nearest endpoint. For each colour the resulting support digraph is acyclic with out-degree at most one, giving at most $n-1$ interior centres, or at most $n-2$ if that colour also has a vertex centre. Together with a local gap lemma, this yields the inequality $\binom{n-c}{2} \leq c(c-2)$ for every feasible c . For the upper bound we reduce the geometric construction to a constrained forest partition. After selecting c neutral roots and reserving one leaf edge for every remaining vertex, a residual graph must be partitioned into c forests with prescribed forbidden vertex sets. We prove the required partition by Edmonds' matroid union theorem. The rank inequalities are reduced, via a distance-two chord completion lemma, to explicit inequalities on induced subgraphs of a graph consisting of a clique joined almost completely to an independent set. A simple level assignment on the resulting forests then gives pairwise disjoint same-colour radius- $\frac{1}{2}$ balls that cover every edge. In particular $\chi_{\frac{1}{2}}(\Gamma(K_{11})) = 5$, whereas the conjectured value is 6, and asymptotically $\chi_{\frac{1}{2}}(\Gamma(K_n)) = (\sqrt{2} - 1)n + O(1)$.

1 Introduction

Replace every edge of a finite graph by a unit interval and identify interval endpoints according to incidence. The result is a one-dimensional geodesic metric space, often called the metric or continuous realization of the graph. Allowing optimization objects to occupy arbitrary points of these intervals produces continuous analogues of classical graph problems.

Grigoriev and Faulkner [1] recently formulated a systematic collection of such problems. For colouring, a continuous graph Γ is (r, c) -colourable if it can be covered by closed radius- r balls whose balls can be assigned c colours so that intersecting balls receive different colours. They write $\chi_r(\Gamma)$ for the least such c and focus in particular on $r = \frac{1}{2}$.

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For the continuous complete graph $\Gamma(K_n)$ they conjectured

$$\chi_{\frac{1}{2}}(\Gamma(K_n)) = \left\lceil \frac{n}{2} \right\rceil, \quad (1)$$

and proved the corresponding upper bound. The missing part was a matching lower bound.

The main theorem of this paper gives the exact answer and shows that the proposed upper bound is generally not tight.

Theorem 1.1 (Main theorem). *For every integer $n \geq 5$, define*

$$\lambda(n) = \min \left\{ c \in \mathbb{Z} : 3 \leq c \leq n, \binom{n-c}{2} \leq c(c-2) \right\}. \quad (2)$$

Then

$$\chi_{\frac{1}{2}}(\Gamma(K_n)) = \lambda(n) = \left\lceil \frac{\sqrt{8n^2 - 24n + 25} - (2n - 5)}{2} \right\rceil. \quad (3)$$

The lower and upper bounds reveal two complementary structures. The lower bound turns each colour class into an oriented forest: every interior ball centre chooses a nearest endpoint of its support edge as its tail. Same-colour disjointness forces distinct tails and forbids directed cycles. The upper bound reverses this viewpoint. We construct suitable forests first, then place ball centres at carefully increasing distances along their oriented edges.

The nontrivial part of the upper construction is therefore combinatorial rather than metric. Choose $c = \lambda(n)$ neutral roots. Every other vertex must receive a nearby “witness” centre, which we force to be a leaf in one colour forest. After reserving those leaf edges, the remaining edges are partitioned into c constrained forests. We prove the partition using the matroid union theorem of Edmonds [2]. The rank verification is the main technical step.

The first failure of (1) occurs at $n = 11$:

$$\chi_{\frac{1}{2}}(\Gamma(K_{11})) = 5 < 6.$$

For odd $n \geq 11$ the conjectured value is already too large, and for even $n \geq 18$ it is too large as well. Thus the conjecture fails for every $n \geq 17$. The exact formula has asymptotic slope $\sqrt{2} - 1 \approx 0.4142$, rather than $1/2$.

2 Continuous colouring and metric preliminaries

Let K_n have original vertex set V . Its continuous realization $\Gamma(K_n)$ is obtained by replacing every edge by a closed unit interval. We use the induced path metric d .

For $p \in \Gamma(K_n)$ and $r \geq 0$, let

$$B(p, r) = \{x \in \Gamma(K_n) : d(p, x) \leq r\}.$$

Definition 2.1. A radius- $\frac{1}{2}$ c -colouring of $\Gamma(K_n)$ is a set of closed balls $B(p, \frac{1}{2})$ that covers $\Gamma(K_n)$, together with a map from the balls to c colours, such that two distinct intersecting balls have different colours. The minimum feasible c is $\chi_{\frac{1}{2}}(\Gamma(K_n))$.

A centre at an original vertex is called a *neutral centre*; a centre in the relative interior of an edge is an *interior centre*. The unique edge containing an interior centre in its relative interior is its *support edge*.

We repeatedly use the following elementary geodesic observation.

Lemma 2.2 (Same-colour centre separation). *Two distinct ball centres of the same colour are at distance strictly greater than 1.*

Proof. If two centres p, q satisfy $d(p, q) \leq 1$, choose a midpoint x of a shortest p - q geodesic. Then $d(p, x), d(q, x) \leq \frac{1}{2}$, so the two closed radius- $\frac{1}{2}$ balls intersect at x . $\square$

Lemma 2.3 (Automatic finiteness). *Every radius- $\frac{1}{2}$ c -colouring of $\Gamma(K_n)$ uses at most cn balls.*

Proof. Every point of a unit edge is within distance $\frac{1}{2}$ of at least one endpoint. Choose, for every ball centre p , an original vertex $\nu(p)$ with $d(p, \nu(p)) \leq \frac{1}{2}$. The map

$$p \mapsto (\text{colour}(p), \nu(p))$$

is injective: if two same-colour centres chose the same vertex v , both balls would contain v . There are only cn colour-vertex pairs. $\square$

Thus the fact that Definition 2.1 permits an arbitrary set of balls causes no compactness issue: every feasible colouring is automatically finite.

3 The general lower bound

Fix a colour. For every interior centre p on an edge xy , choose an endpoint $\tau(p)$ satisfying $d(p, \tau(p)) \leq \frac{1}{2}$, and orient the support edge from $\tau(p)$ to the other endpoint. If p is exactly the midpoint, choose either endpoint. Write $t(p) = d(p, \tau(p)) \leq \frac{1}{2}$.

Lemma 3.1 (One colour supports a forest). *For a fixed colour, the oriented support graph of its interior centres has out-degree at most one and contains no directed cycle. Consequently it has at most $n - 1$ interior centres.*

Proof. Two same-colour interior centres cannot have the same tail vertex, because both radius- $\frac{1}{2}$ balls would contain that vertex. Hence every vertex has out-degree at most one.

Suppose a directed cycle exists,

$$v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_k \rightarrow v_1,$$

and let $t_i \leq \frac{1}{2}$ be the distance of the centre on $v_i v_{i+1}$ from v_i (indices modulo k). For consecutive support edges, the route through the shared vertex v_{i+1} has length

$$(1 - t_i) + t_{i+1}.$$

By Lemma 2.2, the actual centre distance is > 1 ; therefore every particular route between the centres also has length > 1 , and hence $t_{i+1} > t_i$. Going around the cycle gives

$$t_1 < t_2 < \cdots < t_k < t_1,$$

a contradiction.

A finite directed graph with out-degree at most one and no directed cycle has at most $n - 1$ edges. $\square$

Lemma 3.2 (A neutral centre reduces capacity). *If a colour has a neutral centre, then that colour has at most $n - 2$ interior centres.*

Proof. Let the neutral centre be r . No same-colour interior centre can lie on an edge incident with r , because its distance from r is strictly less than 1, contradicting Lemma 2.2. Thus the support forest of Lemma 3.1 lies on at most $n - 1$ original vertices and has at most $n - 2$ edges. $\square$

The second ingredient is a local coverage fact.

Lemma 3.3 (Gap lemma). *If an edge uv contains no interior centre in a radius- $\frac{1}{2}$ colouring, then both u and v are neutral centres.*

Proof. Let m be the midpoint of uv . Any centre whose ball contains m must be u , v , or an interior point of uv : a centre elsewhere must first reach u or v and then travel another distance $\frac{1}{2}$, so its distance from m is $> \frac{1}{2}$. Since there is no interior centre on uv , coverage of m implies that at least one endpoint, say u , is a centre.

Assume v is not a centre. By Lemma 2.3 there are finitely many centres. Every centre $q \neq u$ has $d(q, m) > \frac{1}{2}$. If there is no centre other than u , choose any $\eta \in (0, \frac{1}{2})$. Otherwise choose $\eta > 0$ smaller than the minimum of the finitely many positive gaps $d(q, m) - \frac{1}{2}$ over all centres $q \neq u$. Let x lie on the v -side of m with $0 < d(m, x) < \eta$. The ball at u misses x , while for every other centre

$$d(q, x) \geq d(q, m) - d(m, x) > \frac{1}{2}.$$

Thus x is uncovered, a contradiction. Hence v is also a centre. $\square$

Lemma 3.4 (Neutral centres have distinct colours). *Distinct neutral centres have distinct colours. In particular, if a c -colouring has s neutral centres, then $s \leq c$.*

Proof. For distinct vertices u, v , the balls $B(u, \frac{1}{2})$ and $B(v, \frac{1}{2})$ meet at the midpoint of the edge uv . $\square$

We can now derive the sharp numerical obstruction. We first dispose of the two small colour counts that are not covered by the monotonic elimination used below.

Lemma 3.5 (One and two colours are impossible). *For every $n \geq 5$, $\Gamma(K_n)$ is not radius- $\frac{1}{2}$ c -colourable for $c \leq 2$.*

Proof. Suppose such a colouring exists, and let s be the number of neutral centres. By Lemma 3.4, $s \leq c \leq 2$. By Lemmas 3.1 and 3.2, if M denotes the number of interior centres, then

$$M \leq c(n-1) - s.$$

By Lemma 3.3, every edge without an interior centre has both endpoints among the s neutral vertices, so

$$\binom{n}{2} - \binom{s}{2} \leq M.$$

Consequently

$$\binom{n}{2} \leq c(n-1) + \binom{s}{2} - s \leq 2(n-1),$$

because $\binom{s}{2} - s \leq 0$ for $s = 0, 1, 2$. But

$$\binom{n}{2} - 2(n-1) = \frac{(n-1)(n-4)}{2} > 0$$

for every $n \geq 5$, a contradiction. $\square$

Theorem 3.6 (Lower-bound inequality). *If $\Gamma(K_n)$ admits a radius- $\frac{1}{2}$ c -colouring with $c \geq 3$, then*

$$\binom{n-c}{2} \leq c(c-2). \quad (4)$$

Proof. Let s be the number of neutral centres. By Lemma 3.4, $s \leq c$, and exactly s colours contain a neutral centre.

By Lemmas 3.1 and 3.2, the total number M of interior centres satisfies

$$M \leq s(n-2) + (c-s)(n-1) = c(n-1) - s. \quad (5)$$

By Lemma 3.3, any edge without an interior centre has both endpoints among the s neutral vertices. Hence at least

$$\binom{n}{2} - \binom{s}{2}$$

edges contain an interior centre. Since an interior centre belongs to a unique support edge,

$$\binom{n}{2} - \binom{s}{2} \leq M. \quad (6)$$

Combining (5) and (6) gives

$$\binom{n}{2} \leq c(n-1) + \binom{s}{2} - s.$$

For $0 \leq s \leq c$ and $c \geq 3$, the quantity $g(s) = \binom{s}{2} - s = s(s-3)/2$ is at most $g(c) = c(c-3)/2$. Indeed, $g(0) = 0$, $g(1) = g(2) = -1$, while $g(k+1) - g(k) = k - 1 \geq 1$ for integers $k \geq 2$, and $g(c) \geq 0$. Therefore

$$\binom{n}{2} \leq c(n-1) + \frac{c(c-3)}{2}.$$

Rearranging is exactly (4). □

Corollary 3.7. *For every $n \geq 5$,*

$$\chi_{\frac{1}{2}}(\Gamma(K_n)) \geq \lambda(n).$$

Proof. By Lemma 3.5, every feasible colouring uses at least three colours. Hence any feasible colour count c satisfies the hypothesis of Theorem 3.6, and therefore satisfies (4). By the definition of $\lambda(n)$, this implies $c \geq \lambda(n)$. □

4 The upper bound as a constrained forest decomposition

We now prove the reverse inequality. The cases $5 \leq n \leq 8$ follow immediately from the upper bound $\chi_{\frac{1}{2}}(\Gamma(K_n)) \leq \lceil n/2 \rceil$ of Grigoriev and Faulkner [1], because direct substitution gives

$$\lambda(5) = 3, \quad \lambda(6) = 3, \quad \lambda(7) = 4, \quad \lambda(8) = 4.$$

Hence assume $n \geq 9$ and set

$$c = \lambda(n), \quad u = n - c.$$

Then $c \geq 5$: indeed, $c = 3$ would require $\binom{n-3}{2} \leq 3$, and $c = 4$ would require $\binom{n-4}{2} \leq 8$, both impossible for $n \geq 9$. By definition,

$$\binom{u}{2} \leq c(c-2). \quad (7)$$

Moreover $u \leq 2c$: if $u \geq 2c + 1$, then

$$\binom{u}{2} \geq \binom{2c+1}{2} = c(2c+1) > c(c-2),$$

contradicting (7).

Choose c original vertices

$$R = \{r_1, \dots, r_c\}$$

as *roots*; the other u vertices form a set U . Partition

$$U = W_1 \sqcup \dots \sqcup W_c, \quad |W_i| \leq 2,$$

which is possible because $u \leq 2c$. Indices on the roots are read cyclically modulo c .

For each $v \in W_i$, reserve the edge

$$e_v = vr_{i+1}. \tag{8}$$

Let G be the graph on $R \cup U$ obtained from K_n by deleting all root–root edges and all reserved edges. Thus:

1. U induces a clique;
2. R is independent;
3. every $v \in U$ has exactly one non-neighbour in R , its designated root from (8);
4. every root has at most two non-neighbours in U .

For each i , set

$$Z_i = \{r_i\} \cup W_i.$$

The sets $Z_1, \dots, Z_c$ partition V .

Our combinatorial target is the following.

Proposition 4.1 (Constrained forest partition). *The edge set $E(G)$ can be partitioned*

$$E(G) = I_1 \sqcup \dots \sqcup I_c$$

so that I_i is a forest and no edge of I_i is incident with any vertex of Z_i .

Once this is proved, define

$$F_i = I_i \cup \{vr_{i+1} : v \in W_i\}. \tag{9}$$

Because I_i avoids W_i , every reserved edge in (9) attaches a new leaf. Hence F_i is a forest, it contains no edge incident with r_i , and every $v \in W_i$ is a leaf of F_i . The forests $F_1, \dots, F_c$ partition exactly the edges of K_n that are not root–root edges.

4.1 Matroid formulation

We use the following standard consequence of Edmonds' matroid union theorem [2].

Theorem 4.2 (Matroid union criterion). *Let $M_1, \dots, M_c$ be matroids on a common finite ground set E , with rank functions $r_1, \dots, r_c$. Then E can be partitioned as $E = I_1 \sqcup \dots \sqcup I_c$ with I_i independent in M_i if and only if*

$$|A| \leq \sum_{i=1}^c r_i(A) \quad \text{for every } A \subseteq E. \tag{10}$$

For each i , let M_i be the graphic matroid of $G - Z_i$, extended to the ground set $E(G)$ by making every edge incident with Z_i a loop. Thus the independent sets of M_i are precisely the forests using only vertices outside Z_i .

It remains to verify (10).

4.2 The rank formula

It is enough to consider a connected edge set $C \subseteq E(G)$: both $|C|$ and each graphic rank are additive over the connected components of an arbitrary edge set.

Let $W = V(C)$ and $w = |W|$. For each i , let $\kappa_i(C)$ be the number of connected components of $C - Z_i$ on the full surviving vertex set $W \setminus Z_i$, counting isolated surviving vertices; set $\kappa_i(C) = 0$ if $W \setminus Z_i = \emptyset$. Define

$$Q(C) = \sum_{i=1}^c (\kappa_i(C) - 1). \quad (11)$$

Because the Z_i partition the vertex set,

$$\sum_i |W \setminus Z_i| = cw - w.$$

The graphic rank in M_i is $|W \setminus Z_i| - \kappa_i(C)$, so

$$\sum_{i=1}^c r_i(C) = c(w - 1) - w - Q(C). \quad (12)$$

Thus the desired rank inequality is equivalent to

$$|E(C)| + Q(C) \leq c(w - 1) - w. \quad (13)$$

4.3 Completion by distance-two chords

The special form of G allows us to reduce (13) from arbitrary connected subgraphs to induced subgraphs.

Lemma 4.3 (Distance-two chord). *Let C be a connected proper spanning subgraph of $G[W]$. Then there exist vertices $x, z, y \in W$ such that $xz, zy \in E(C)$ and $xy \in E(G[W]) \setminus E(C)$.*

Proof. Choose a missing allowed edge $xy \in E(G[W]) \setminus E(C)$ and a shortest x - y path

$$x = v_0, v_1, \dots, v_k = y$$

in C . If some distance-two pair $v_j v_{j+2}$ is an edge of G , then it is not an edge of C (otherwise the path was not shortest), and we are done. Suppose therefore that every distance-two pair on the path is a non-edge of G .

Classify vertices as roots (R) or ordinary vertices (U). Consecutive path vertices cannot both be roots because R is independent. A distance-two pair cannot both be ordinary because U is a clique. Hence any four consecutive vertices must have type

$$R - U - U - R;$$

this is the unique length-four R/U pattern satisfying those two restrictions.

If $k = 3$, the endpoints x, y are both roots, contradicting $xy \in E(G)$. If $k \geq 4$, the first four vertices force the pattern $R - U - U - R$, while the next four vertices would also have to be $R - U - U - R$, forcing v_1 to be both U and R . This contradiction proves the claim. (The case $k = 2$ gives the desired chord immediately.) $\square$

Lemma 4.4 (Completion monotonicity). *Let C' be obtained from a connected C by adding a chord xy from Lemma 4.3, with $x - z - y$ the corresponding length-two path in C . Then*

$$|E(C')| + Q(C') \geq |E(C)| + Q(C).$$

Proof. Adding the chord increases the edge count by one. For every i with $z \notin Z_i$, if x and y survive deletion of Z_i , then the path $x - z - y$ also survives, so x and y were already in the same component of $C - Z_i$. Hence κ_i does not decrease. The vertex z lies in exactly one set Z_j ; only for this single index can the new edge merge two components, and then κ_j decreases by at most one. Thus $Q(C') \geq Q(C) - 1$. $\square$

Repeatedly applying Lemmas 4.3 and 4.4, we obtain:

Corollary 4.5. *For a fixed vertex set W with $G[W]$ connected, the quantity $|E(C)| + Q(C)$ is maximized, among connected spanning subgraphs $C \subseteq G[W]$, by $G[W]$ itself. Hence it suffices to verify (13) for induced graphs $G[W]$.*

4.4 Checking the induced subgraphs

Fix $W \subseteq V$ for which $H = G[W]$ is connected. Put

$$a = |W \cap R|, \quad b = |W \cap U|,$$

so $w = a + b$. Let m be the number of deleted reserved pairs with both endpoints in W . Then

$$|E(H)| = \binom{b}{2} + ab - m. \quad (14)$$

Each root outside $W \cap R$ is the designated non-neighbour of at most two ordinary vertices. Therefore at most $2(c - a)$ of the b ordinary vertices can have their designated root outside W , and hence

$$m \geq 2a + b - 2c. \quad (15)$$

(The bound is harmless when its right-hand side is negative.)

We next control $Q(H)$. Write $b_i = |W \cap W_i| \leq 2$. After deleting Z_i , the number of surviving ordinary vertices is $b - b_i$. Those vertices form a clique. If at least three survive, every surviving root is adjacent to at least one of them because a root has at most two non-neighbours in U . Therefore

$$b \geq 5 \implies Q(H) = 0. \quad (16)$$

For $b \leq 4$, a short case analysis gives

$$\begin{array}{c|cccc} b & 1 & 2 & 3 & 4 \\ \hline Q(H) & \leq a - 1 & \leq a + 1 & \leq 2 & \leq 2. \end{array} \quad (17)$$

For completeness, the case analysis is given in section A.

Define the slack in (13) by

$$S = c(a + b - 1) - (a + b) - |E(H)| - Q(H).$$

We prove $S \geq 0$.

First suppose $b \geq 5$. By (16), (14), and (15),

$$S \geq a(c + 1 - b) + c(b - 3) - \binom{b}{2} =: D(a, b). \quad (18)$$

If $b \leq c + 1$, then the first term in D is nonnegative. Also

$$c(b - 3) - \binom{b}{2} \geq 0 :$$

for $b = 5, 6$ this follows from $c \geq 5$, while for $b \geq 7$ we have $c \geq b - 1$ and

$$2 \left(c(b-3) - \binom{b}{2} \right) \geq (b-1)(b-6) \geq 0.$$

Thus $D \geq 0$.

If $b > c + 1$, the coefficient of a in (18) is negative, so D is minimized at $a = c$. Hence

$$D(a, b) \geq D(c, b) = c(c-2) - \binom{b}{2} \geq c(c-2) - \binom{u}{2} \geq 0$$

by $b \leq u$ and (7).

It remains to check $1 \leq b \leq 4$. Using only $m \geq 0$ and the bounds (17), direct simplification gives

$$\begin{aligned} b = 1 : \quad S &\geq a(c-3), \\ b = 2 : \quad S &\geq (a+1)(c-4), \\ b = 3 : \quad S &\geq (a+2)(c-4), \\ b = 4 : \quad S &\geq a(c-5) + 3(c-4). \end{aligned}$$

All are nonnegative because $c \geq 5$. The case $b = 0$ contains no edge because R is independent and therefore does not arise from a nonempty connected edge set.

We have proved (13) for every connected edge set. By Theorem 4.2, Proposition 4.1 follows.

5 From forests to a continuous colouring

We now turn the forest partition into actual ball centres.

Lemma 5.1 (Geometric realization of the forests). *Let $F_1, \dots, F_c$ be the forests constructed in (9). Then $\Gamma(K_n)$ admits a radius- $\frac{1}{2}$ c -colouring.*

Proof. For colour i , place a neutral centre at r_i . Consider a component of F_i . If it contains one of the prescribed leaves W_i , choose its sink to be a vertex outside W_i ; this is always possible because each such leaf is adjacent to r_{i+1} . Otherwise choose any sink. Orient every edge of the tree toward its sink. Thus every vertex has out-degree at most one, and each prescribed leaf $v \in W_i$ is oriented as

$$v \rightarrow r_{i+1}.$$

Assign positive integer levels to vertices so that every source has level 1 and levels strictly increase along every oriented edge. For instance, set

$$\ell(x) = 1 + \max\{\text{length of a directed path from a source to } x\}.$$

Every tail of an oriented edge has $1 \leq \ell(x) \leq n-1$.

Set

$$\varepsilon = \frac{1}{2n}.$$

For every oriented edge $x \rightarrow y$ of F_i , place a colour- i interior centre on xy at distance

$$t_x = \ell(x)\varepsilon$$

from x . Then

$$\varepsilon \leq t_x < \frac{1}{2},$$

and on every oriented path $x \rightarrow y \rightarrow z$ we have $t_y > t_x$. In particular, because every $v \in W_i$ is a source leaf, its reserved edge carries a centre exactly ε from v .

We first prove same-colour disjointness. The neutral centre r_i is not incident with any support edge of F_i , so its distance to every interior centre of colour i is > 1 . For two interior centres, their support edges are distinct. If the supports are disjoint, every endpoint route includes a unit edge between distinct original vertices plus two positive offsets, and hence has length > 1 . If the supports share a vertex, the orientations can only be of the forms

$$x \rightarrow v \rightarrow y \quad \text{or} \quad x \rightarrow v \leftarrow y,$$

because every vertex has out-degree at most one. In the first case the route through v has length

$$(1 - t_x) + t_v > 1$$

since $t_v > t_x$; all other endpoint routes contain a full unit edge and positive additional length. In the second case the route through v has length

$$(1 - t_x) + (1 - t_y) > 1$$

since $t_x, t_y < \frac{1}{2}$, and again the other routes are > 1 . Therefore every pair of distinct same-colour centres has distance > 1 . If two of their radius- $\frac{1}{2}$ balls intersected, the triangle inequality would force the centre distance to be at most 1; hence the balls are disjoint.

It remains to prove coverage. Every root-root edge is covered by the two neutral balls at its endpoints. Now let xy be any other edge. It belongs to exactly one forest, say F_i , and is oriented $x \rightarrow y$. Its own centre lies at distance $t_x \in [\varepsilon, \frac{1}{2})$ from x , so its radius- $\frac{1}{2}$ ball covers the segment of xy from x through coordinate $\frac{1}{2} + t_x$, measured from x .

If y is a root, its neutral ball covers the final half of xy , and the two covered intervals overlap. If $y \in U$, write $y \in W_j$. The reserved edge of colour j carries a centre exactly ε from y . A point of xy at coordinate s from x is at distance at most

$$\varepsilon + (1 - s)$$

from this witness centre via y , so the witness ball covers every point with $s \geq \frac{1}{2} + \varepsilon$. Since $t_x \geq \varepsilon$, the ball on xy itself reaches at least $\frac{1}{2} + \varepsilon$. Hence the two balls cover the entire edge. Thus all of $\Gamma(K_n)$ is covered. $\square$

Theorem 5.2 (Upper bound). *For every $n \geq 5$,*

$$\chi_{\frac{1}{2}}(\Gamma(K_n)) \leq \lambda(n).$$

Proof. For $5 \leq n \leq 8$, use the known upper bound $\lceil n/2 \rceil = \lambda(n)$. For $n \geq 9$, Proposition 4.1 and Lemma 5.1 construct a c -colouring with $c = \lambda(n)$. $\square$

Combining Corollary 3.7 and Theorem 5.2 proves the first equality in Theorem 1.1.

6 Closed form and consequences

The inequality in (2) is equivalent to

$$c^2 + (2n - 5)c - (n^2 - n) \geq 0.$$

The relevant root of the quadratic is

$$\rho_n = \frac{\sqrt{8n^2 - 24n + 25} - (2n - 5)}{2}.$$

For $n \geq 5$, the quadratic is negative at $c = 2$ and positive at $c = n$, since

$$2^2 + (2n - 5)2 - (n^2 - n) = -(n - 2)(n - 3) < 0$$

and

$$n^2 + (2n - 5)n - (n^2 - n) = 2n(n - 2) > 0.$$

Hence its positive root satisfies $2 < \rho_n < n$, so the constraint $3 \leq c \leq n$ in (2) is automatic for the least feasible integer. Therefore that least integer is $\lceil \rho_n \rceil$, proving the second equality in Theorem 1.1.

Corollary 6.1 (First values). *For $5 \leq n \leq 20$ the exact values are*

n	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$\chi_{\frac{1}{2}}(\Gamma(K_n))$	3	3	4	4	5	5	5	6	6	7	7	8	8	8	9	9.

In particular K_{11} is the first counterexample to (1).

Proof. The values follow by substitution. For $5 \leq n \leq 10$, they coincide with $\lceil n/2 \rceil$. At $n = 11$ the formula gives 5 rather than 6. $\square$

Corollary 6.2 (Infinite failure of the conjecture). *For every odd $n \geq 11$ and every even $n \geq 18$,*

$$\chi_{\frac{1}{2}}(\Gamma(K_n)) < \left\lceil \frac{n}{2} \right\rceil.$$

Consequently the conjecture (1) fails for every $n \geq 17$.

Proof. If $n = 2m + 1$ with $m \geq 5$, take $c = m = \lceil n/2 \rceil - 1$. Then $n - c = m + 1$ and

$$\binom{m+1}{2} \leq m(m-2)$$

is equivalent to $m \geq 5$. Thus $\lambda(n) \leq m$.

If $n = 2m$ with $m \geq 9$, take $c = m - 1$. The defining inequality becomes

$$\binom{m+1}{2} \leq (m-1)(m-3),$$

equivalent to $m^2 - 9m + 6 \geq 0$, which holds for $m \geq 9$. $\square$

Corollary 6.3 (Asymptotics). *As $n \rightarrow \infty$,*

$$\chi_{\frac{1}{2}}(\Gamma(K_n)) = (\sqrt{2} - 1)n + O(1).$$

7 The explicit K_{11} certificate

The general proof is constructive through a matroid partition, but the first counterexample also has a compact exact rational certificate. We record it because it is useful for independent checking and for computational exploration.

Use colours 0, 1, 2, 3, 4. Put a neutral centre of colour i at vertex i . An entry $u \rightarrow v : k$ denotes the point at distance $k/13$ from u on the edge uv .

Exact rational calculation gives minimum same-colour centre distance $14/13 > 1$ in each colour and verifies continuous coverage of all 55 unit edges. This certificate is not needed for the proof of Theorem 1.1; it is an explicit small witness for the first counterexample.

Colour	Neutral	Interior centres $u \rightarrow v : k$
0	0	$1 \rightarrow 6 : 1, 2 \rightarrow 6 : 6, 9 \rightarrow 2 : 5, 3 \rightarrow 9 : 1, 7 \rightarrow 9 : 4, 8 \rightarrow 7 : 3, 10 \rightarrow 8 : 2, 4 \rightarrow 10 : 1, 5 \rightarrow 10 : 1$
1	1	$0 \rightarrow 5 : 4, 3 \rightarrow 5 : 1, 4 \rightarrow 5 : 1, 7 \rightarrow 0 : 2, 9 \rightarrow 0 : 3, 6 \rightarrow 7 : 1, 8 \rightarrow 9 : 1, 10 \rightarrow 9 : 2, 2 \rightarrow 10 : 1$
2	2	$6 \rightarrow 3 : 2, 7 \rightarrow 3 : 3, 10 \rightarrow 3 : 1, 0 \rightarrow 6 : 1, 5 \rightarrow 6 : 1, 9 \rightarrow 6 : 1, 1 \rightarrow 7 : 1, 4 \rightarrow 7 : 2, 8 \rightarrow 4 : 1$
3	3	$5 \rightarrow 2 : 5, 1 \rightarrow 5 : 1, 7 \rightarrow 5 : 3, 9 \rightarrow 5 : 4, 10 \rightarrow 7 : 2, 4 \rightarrow 9 : 3, 0 \rightarrow 10 : 1, 6 \rightarrow 4 : 2, 8 \rightarrow 6 : 1$
4	4	$1 \rightarrow 9 : 4, 8 \rightarrow 1 : 3, 10 \rightarrow 1 : 2, 0 \rightarrow 8 : 1, 2 \rightarrow 8 : 2, 3 \rightarrow 8 : 1, 5 \rightarrow 8 : 1, 6 \rightarrow 10 : 1, 7 \rightarrow 2 : 1$

Table 1: An exact five-colour certificate for $\Gamma(K_{11})$.

8 Discussion

The formula in Theorem 1.1 changes the picture suggested by the matching-based upper bound $\lceil n/2 \rceil$. Optimal colour classes are naturally governed by forests rather than matchings. The lower bound discovers this structure from metric separation, while the upper bound deliberately builds the same structure and adds one prescribed leaf per ordinary vertex to guarantee coverage.

The coefficient $\sqrt{2} - 1$ is forced by balancing two quantities. If c vertices are neutral roots, then roughly $\binom{n-c}{2}$ edges remain to be supported among ordinary vertices, while the c colours have total forest capacity on the order of c^2 . The exact inequality

$$\binom{n-c}{2} \leq c(c-2)$$

is therefore both the lower-bound obstruction and the upper-bound feasibility condition.

The constrained forest partition in Proposition 4.1 may be of independent interest. Its host graph is a split-like graph with a clique U , an independent set R , and a degree-defect pattern in which every U -vertex misses exactly one root and every root misses at most two U -vertices. The distance-two chord lemma is what makes the full matroid-union rank test tractable: it shows that the worst connected edge set on a fixed vertex set is the induced host itself.

Several directions remain open. One may ask for analogous formulas for other radii, for complete multipartite or other dense graph families, or for algorithmic versions of the constrained forest construction. The general behaviour of $\chi_r(\Gamma(G))$ as a function of r also appears largely unexplored.

Research process and AI assistance

This work originated in an exploratory workflow that the author informally calls *vibe mathing*: the human author selected the open problem, specified the research target, iteratively challenged proposed arguments, requested adversarial checks, ran exact computations and formal builds, and curated the resulting manuscript. The central proof strategy and a substantial majority of the proof discovery and initial proof development were generated by OpenAI’s GPT-5.6 Sol under the author’s direction. In particular, the AI system proposed the candidate exact formula, developed the orientation/forest lower-bound argument, and developed the constrained-forest/matroid-union route used for the upper bound. The human author performed the research steering, selected which arguments to retain, identified and corrected gaps through repeated checking, executed the computational and formal verification workflow, and assumes responsibility for every mathematical claim in the submitted manuscript.

This disclosure concerns the provenance of the research, not its evidentiary basis. No statement in the paper is justified by an appeal to AI output. The general theorem is intended to stand entirely on the self-contained mathematical proof given above, and the author particularly welcomes independent scrutiny of the matroid-union and geometric-realization arguments. The AI system is not listed as an author and does not bear responsibility for the work.

Supplementary verification

The mathematical proof of Theorem 1.1 does not depend on computer-assisted verification. As an independent check of the first counterexample, the K_{11} specialization has been verified by exact rational arithmetic and formalized in Lean 4/mathlib. A complete Lean project has built successfully without `sorry` or user-introduced axioms; some finite exact certificate checks in the currently frozen verification project use Lean’s `native_decide` mechanism. These artifacts verify the K_{11} specialization only, not the general theorem. A reproducibility archive can be supplied to reviewers and will be deposited publicly with versioned source and build instructions before final publication.

A The small- b component bounds

We justify (17). Recall that $b_i = |W \cap W_i| \leq 2$ and that after deleting Z_i there remain $t_i = b - b_i$ ordinary vertices. They form a clique. If $t_i = 2$, at most one surviving root can be isolated, and this can happen only when the two surviving ordinary vertices belong to the same block W_j ; the only possible isolated root is their common designated non-neighbour. If $t_i = 1$, at most the unique designated non-neighbour of that ordinary vertex can be isolated. If $t_i = 0$, all surviving roots are isolated.

$b = 1$. The unique ordinary vertex lies in some block W_j . Because H is connected, its designated non-neighbour is not among the a selected roots. For $i = j$, deleting W_j leaves only roots, contributing at most $a - 1$ to $\kappa_i - 1$. For every other i , the ordinary vertex survives and is adjacent to every surviving root, so the contribution is zero. Hence $Q \leq a - 1$.

$b = 2$. If the two ordinary vertices lie in the same block W_j , connectedness excludes their common designated non-neighbour from $W \cap R$. Deleting Z_j contributes at most $a - 1$, and every other deletion leaves a connected graph; hence $Q \leq a - 1$. If they lie in two distinct blocks, deleting either occupied block leaves one ordinary vertex and contributes at most one; every other deletion leaves two ordinary vertices from distinct blocks and hence a connected graph. Thus $Q \leq 2$, which is at most $a + 1$ when $a \geq 1$, while for $a = 0$ all contributions are actually zero. Therefore $Q \leq a + 1$ in all cases.

$b = 3$. If the three vertices lie in three distinct blocks, deleting any one block leaves two ordinary vertices from different blocks, so no root is isolated and $Q = 0$. If the block pattern is $2 + 1$, deleting the double block leaves one ordinary vertex and contributes at most one, while deleting the single block leaves the two vertices of the double block and contributes at most one. All other deletions leave at least three ordinary vertices. Hence $Q \leq 2$.

$b = 4$. Only an index with $b_i = 2$ can leave fewer than three ordinary vertices. Each such index contributes at most one, and there are at most two such indices. Therefore $Q \leq 2$.

B Machine-readable K_{11} certificate

For reproducibility, an interior triple (u, v, k) denotes the centre at coordinate $k/13$ from u on edge uv ; colour i additionally has neutral centre i .

0: (1,6,1) (2,6,6) (9,2,5) (3,9,1) (7,9,4)
 (8,7,3) (10,8,2) (4,10,1) (5,10,1)
1: (0,5,4) (3,5,1) (4,5,1) (7,0,2) (9,0,3)
 (6,7,1) (8,9,1) (10,9,2) (2,10,1)
2: (6,3,2) (7,3,3) (10,3,1) (0,6,1) (5,6,1)
 (9,6,1) (1,7,1) (4,7,2) (8,4,1)
3: (5,2,5) (1,5,1) (7,5,3) (9,5,4) (10,7,2)
 (4,9,3) (0,10,1) (6,4,2) (8,6,1)
4: (1,9,4) (8,1,3) (10,1,2) (0,8,1) (2,8,2)
 (3,8,1) (5,8,1) (6,10,1) (7,2,1)

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